



جامعة بجاية
Tasdawit n Bgayet
Université de Béjaïa

UNIVERSITY OF BÉJAÏA

DEPARTMENT OF COMPUTER SCIENCE

Ergonomics and Accessibility in HCI for People with Disabilities in Algeria

A Literature Review

Prepared by:

Mrs. Asma BACHIR BEY

Supervised by:

Mrs. Karima AIT
ABDELOUHAB

December 17, 2025

Abstract

This literature review examines the intersection of ergonomics, accessibility, and Human-Computer Interaction (HCI) with a focus on the Algerian context. As digital technologies become central to daily life, ensuring equitable access for people with disabilities is a global priority. The paper first establishes a typology of digital interaction barriers—sensory, physical/motor, and cognitive—and links them to core ergonomic principles. It then outlines the foundational international guidelines and standards, such as ISO 9241-210, WCAG, and RGAA, that provide a framework for inclusive design. A survey of assistive and adaptive technologies demonstrates existing solutions for various impairments. Finally, the review examines the Algerian context, showing that despite legal frameworks and a growing population of people with disabilities, digital accessibility remains limited, with no evidence of widespread enforcement of accessibility standards.

List of Figures

1	Figure illustrating Human-Computer Interaction (HCI) <i>Note:</i> Adapted from the concept presented in “HUMAN-COMPUTER INTERACTION (HCI)” by F. Pathan, L. Yadav, & A. Deharkar (2024). Image provided by Pngtree.	4
2	Disability prevalence in Algeria by age, sex, place of residence, and region. Adapted from “The prevalence of disability in Algeria: Reality and Prospects,” by Achedi Khadra and Sarah Bouledras. Data source: MICS4.2012; MICS3.2006.	5
3	Simulation of Age-Related Macular Degeneration (AMD).	6
4	Simulation of vision loss due to diabetic retinopathy.	6
5	Hierarchy of WCAG compliance levels (A, AA, AAA).	9
6	Example of a refreshable Braille display.	11
7	The input bandwidth spectrum for computer interaction, ranging from low bandwidth switches to high bandwidth touch typing.	12
8	Diagram of a computer joystick components: 1. stick, 2. base, 3. trigger, 4. extra buttons, 5. autofire switch, 6. throttle, 7. hat switch (POV hat), 8. suction cups.	12
9	The Kensington Expert Mouse Trackball.	12
10	The Adjustable Head Stylus, an example of a head pointer for hands-free tablet access.	13
11	Example of a head-boxed eye-tracking setup.	13
12	A standard single switch, used as a primary input device for users with limited motor control.	14
13	An adapted keyboard with a non-QWERTY layout.	14
14	Conceptual diagram of a Brain-Computer Interface (BCI) for typing. . . .	15

Contents

1	Introduction	4
2	HCI Accessibility Challenges: A Typology of Interaction Barriers	5
2.1	Sensory Barriers	5
2.1.1	Visual impairments	6
2.1.2	Hearing impairments	7
2.2	Physical and Motor Barriers	7
2.3	Cognitive, Intellectual, and Mental Health Barriers	7
3	Foundational Guidelines and Standards	8
3.1	ISO 9241-210	8
3.2	Web Content Accessibility Guidelines (WCAG)	8
3.2.1	WCAG Conformance Levels	8
3.3	Référentiel Général d'Amélioration de l'Accessibilité (RGAA)	9
4	Assistive and Adaptive Technologies: Solutions for Digital Inclusion	10
4.1	Solutions for Sensory Impairments	10
4.1.1	Visual Impairments	10
4.1.2	Hearing Impairments	11
4.2	Solutions for Physical and Motor Impairments	11
4.2.1	Alternative Pointing Devices	12
4.2.2	Eye-Tracking (Oculometry)	13
4.2.3	Switch Access	13
4.2.4	Speech Input	14
4.2.5	Adapted Keyboards	14
4.2.6	Brain-Computer Interfaces (BCI)	14
4.3	Solutions for Cognitive Barriers	15
5	Digital Accessibility in Algeria: Context and Gaps	15
6	Conclusion	16

1 Introduction

As digital technologies increasingly mediate human experiences across education, employment, healthcare, and social participation, emphasizing accessibility when designing interfaces is paramount since inclusivity greatly impacts how broadly these technologies can be adopted across different populations [1]. Human-Computer Interaction (HCI) is a multidisciplinary field at the intersection of computer science, psychology, design, and other disciplines. Its purpose is to study how humans interact with computers and to design systems that facilitate intuitive and efficient interaction [2]. The goal of universal accessibility is to ensure that this interaction is possible for all individuals, regardless of their abilities, tasks, or technological platforms [3].



Figure 1: Figure illustrating Human-Computer Interaction (HCI)

Note: Adapted from the concept presented in “HUMAN-COMPUTER INTERACTION (HCI)” by F. Pathan, L. Yadav, & A. Deharkar (2024). Image provided by Pngtree.

While international accessibility standards and assistive technologies have advanced significantly, substantial disparities persist between and within nations. In Algeria, this divide overlaps with demographic realities reflecting the breadth of needs and depth of exclusion. According to Multi-Indicator Cluster Survey (MICS) data, disability prevalence in Algeria increased from approximately 1.2% in 2006 to 1.7% in 2013 [4]. The demographic profile shows higher disability rates among males, with elderly men (60+) comprising the most vulnerable group. Southern Algerian regions exhibit elevated disability rates, particularly motor impairments.

These statistics underscore a critical gap: despite Algeria’s growing disabled population, systematic digital barriers persist. As projections indicate continued growth in disability prevalence through 2038, understanding and addressing these HCI accessibility challenges becomes not merely a technological concern but a matter of human rights and social inclusion [4]. This literature review analyzes the barriers in digital interaction, the standards designed to overcome them, the assistive technologies available, and the contextual challenges and opportunities within Algeria.

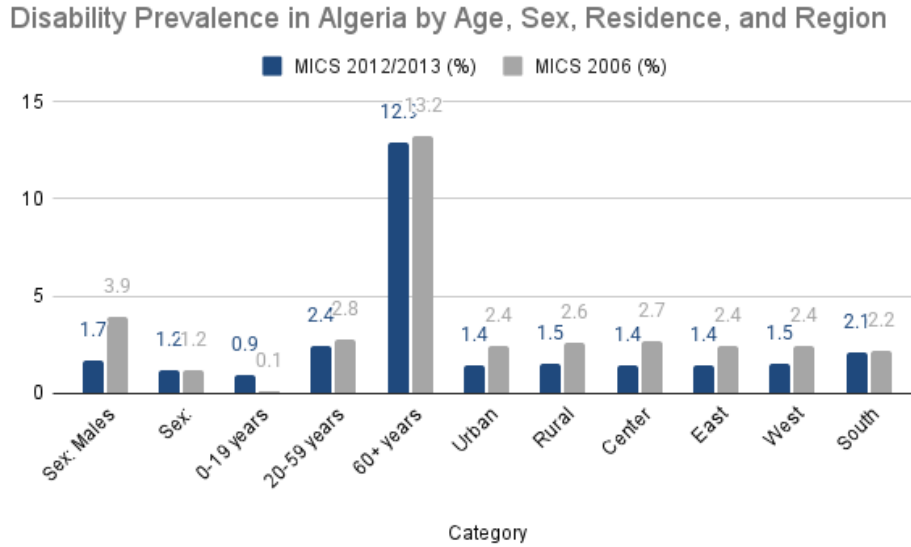


Figure 2: Disability prevalence in Algeria by age, sex, place of residence, and region. Adapted from “The prevalence of disability in Algeria: Reality and Prospects,” by Achedi Khadra and Sarah Bouledras. Data source: MICS4.2012; MICS3.2006.

2 HCI Accessibility Challenges: A Typology of Interaction Barriers

In Human-Computer Interaction (HCI), accessibility issues frequently arise when interactive systems fail to meet the diverse requirements of users with disabilities [5]. This may occur, for example, when the interface does not align with the perceptual, cognitive, or motor abilities of users. According to the social model of disability, as reflected in the UN Convention on the Rights of Persons with Disabilities (UNCRPD), persons with disabilities “include those who have long-term physical, mental, intellectual or sensory impairments which, in interaction with various barriers, may hinder their full and effective participation in society on an equal basis with others” [6]. In practice, these barriers can be broadly categorized into sensory, physical/motor, and cognitive or mental domains, affecting users’ ability to perceive information, provide input, or understand system feedback. Each type of disability influences user–system interaction differently, often affecting one or more of the key ergonomic principles of accessibility, namely: perceivable, operable, understandable, and robust. If any of these principles are not met, users with disabilities may be unable to access or effectively use digital content [7].

2.1 Sensory Barriers

Sensory barriers primarily affect the channels of vision and hearing, which are fundamental for perceiving digital information.

2.1.1 Visual impairments

These encompass a wide spectrum, from partial sight (low vision) and color blindness to complete blindness. Conditions such as glaucoma, age-related macular degeneration (AMD), and diabetic retinopathy hinder the user’s ability to perceive graphical information, text, and visual indicators on screens [8]. This directly undermines the WCAG “Perceivable” principle. For instance, an image without a text alternative (alt text) is completely inaccessible to a blind user relying on a screen reader. Similarly, insufficient color contrast makes text unreadable for users with low vision or color vision deficiencies (e.g., daltonism).



Figure 3: Simulation of Age-Related Macular Degeneration (AMD).

Note. From “Latest advances in treating age-related macular degeneration,” by R. R. Reddy, 2019, Neoretna Eyecare Institute. Retrieved from <https://www.neoretna.com/blog/latest-advances-amd-treatment/>.

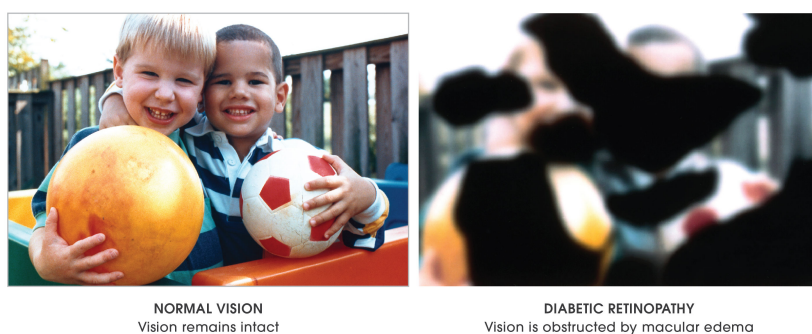


Figure 4: Simulation of vision loss due to diabetic retinopathy.

Note. Adapted from “Quality Analysis of Retina Images for the Automatic Diagnosis of Diabetic Retinopathy,” by L. Giancardo and F. Meriaudeau, National Eye Institute image (Figure 1.2). Image courtesy of the National Eye Institute.

2.1.2 Hearing impairments

Hearing loss, ranging from mild to profound deafness, prevents users from perceiving audio cues, alerts, or verbal feedback in multimedia content [9]. A user may not hear an error notification sound or miss crucial information in a video tutorial without captions. This violates the principle of providing redundant, non-auditory information. The WCAG standard explicitly recommends providing captions for prerecorded and live audio content and other alternatives for time-based media to ensure equivalent access [7].

2.2 Physical and Motor Barriers

Motor disabilities affect the user's ability to manipulate standard input devices. These can include paralysis, tremors, loss of limbs, arthritis, or conditions like cerebral palsy, which reduce precision, range of motion, or make the use of upper limbs unavailable. These conditions make precise cursor movements, double-clicking, or maintaining prolonged touchscreen gestures difficult or impossible. This severely limits "operability," a core WCAG and ergonomic principle. A website that requires a hover action to reveal a menu, for example, is inaccessible to someone who cannot control a mouse with precision. Similarly, interfaces with short timeouts for responses can exclude users who require more time to physically formulate their input via alternative methods [7].

2.3 Cognitive, Intellectual, and Mental Health Barriers

This category represents a diverse set of challenges that affect memory, attention, processing speed, language comprehension, problem-solving skills, and emotional regulation. It includes conditions such as dyslexia, autism spectrum disorder (ASD), attention-deficit/hyperactivity disorder (ADHD), intellectual disabilities, and mental health conditions like anxiety [1, 10]. These conditions can reduce understandability, which is a core principle of accessible and ergonomic design.

Dyslexia: creates a barrier to processing and understanding written text. Users may struggle with dense blocks of text or certain fonts, leading to fatigue and errors [11].

Autism Spectrum Disorder: can cause cognitive overload when interfaces contain unpredictable layouts, flashing elements, or overly complex navigation structures. A tendency toward literal interpretation also means that metaphorical language or inconsistent icons may become confusing [12].

Intellectual disabilities: may introduce difficulties related to cognitive load, memory, and understanding abstract concepts. Multi-step processes or complex forms are often challenging to complete.

Mental health conditions: such as anxiety, can make interaction stressful when interfaces are complicated, time-pressured, or create fear of making irreversible mistakes.

3 Foundational Guidelines and Standards

Several international standards address accessibility barriers, providing actionable frameworks.

3.1 ISO 9241-210

ISO 9241-210: Ergonomics of human-system interaction — Part 210 focuses on human-centred design for interactive systems. Rather than providing a checklist, it establishes an iterative user-centred design (UCD) process that involves users throughout the development lifecycle. The process includes understanding the context of use, specifying user requirements, producing design solutions, and evaluating them against those requirements [13]. This philosophy is essential for accessibility, as it ensures that designers consider the full range of human diversity from the outset.

3.2 Web Content Accessibility Guidelines (WCAG)

Developed by the World Wide Web Consortium (W3C), the WCAG is the globally recognized standard for web accessibility. It is organized around four core principles known as POUR: perceivable, operable, understandable, and robust. These principles ensure that content is accessible to a wide range of users, including those who rely on assistive technologies.

Perceivable: Information and user interface components must be presented in ways that users can perceive (e.g., text alternatives for images, captions for videos) [7].

Operable: User interface components and navigation must be operable (e.g., keyboard accessibility, adjustable time limits) [7].

Understandable: Information and the operation of the interface must be understandable (e.g., readable text, predictable behavior) [7].

Robust: Content must be robust enough to be interpreted reliably by a variety of user agents, including assistive technologies [7].

The latest version, WCAG 2.2, defines three levels of conformance: A (minimum), AA (the most commonly targeted standard), and AAA (the highest level) [14].

3.2.1 WCAG Conformance Levels

The Web Content Accessibility Guidelines define three levels of compliance—A, AA, and AAA—each building on the previous level by introducing additional criteria and stricter requirements [15]. These levels help evaluate how accessible web content is for users with disabilities.

Level A: Minimal Compliance represents the most basic level of accessibility. It addresses essential barriers such as providing text alternatives for non-text content, ensur-

ing keyboard accessibility, and allowing sufficient time for users to read and interact with content [15]. While Level A ensures basic access, it does not guarantee a fully comfortable user experience.

Level AA: Good Compliance builds on Level A and targets a broader range of accessibility issues. Requirements include appropriate text-to-background contrast, the ability to resize text up to 200% without loss of functionality, and the availability of multiple navigation mechanisms [15]. Level AA is generally considered the standard for accessible design.

Level AAA: Exceptional Compliance includes all Level A and AA criteria plus additional measures that maximize accessibility. Examples include enhanced contrast, extended audio descriptions for video, and sign language interpretation [15]. Achieving Level AAA is challenging but provides the highest level of accessibility.

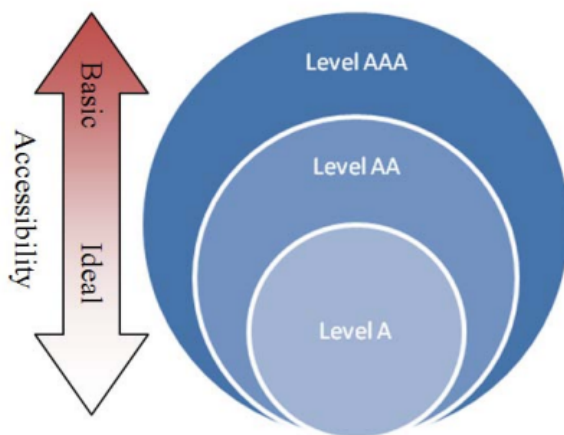


Figure 5: Hierarchy of WCAG compliance levels (A, AA, AAA).

Note. Adapted from “Exploring social media usage in Saudi e-government websites,” by M. Al-Razgan, 2014, **International Journal of Computer Science and Network Security**, 14(3), p. 110 [PDF].

©2014 by the author. Reprinted with permission. Retrieved from

https://www.researchgate.net/profile/Muna-Al-Razgan/publication/262369345_Exploring_social_media_usage_in_Saudi_e-government_websites/links/56f3d03408ae95e8b6cdd9e1/Exploring-social-media-usage-in-Saudi-e-government-websites.pdf

3.3 Référentiel Général d’Amélioration de l’Accessibilité (RGAA)

The RGAA is the French government’s framework for implementing digital accessibility. It is based heavily on the WCAG and adapts the international guidelines to French legal requirements. The RGAA provides a detailed technical reference and a self-assessment methodology. It is organized into 13 thematic groups (e.g., Images, Frames, Colors, Multimedia, Tables) comprising 106 testable criteria, making it a practical tool for both developers and auditors working to ensure compliance with WCAG A and AA [16].

The following sections detail some of the most critical RGAA criteria, illustrating how

they put accessibility into practice.

Criteria 1: Images (Perceivability) ensures that all non-text content is accessible. Informative images require concise, meaningful `alt` text, decorative images must include an empty `alt=""`, and complex graphics require a longer description[16].

Criteria 2: Frames (Operability) requires each frame to have a descriptive `title` attribute that clarifies its purpose and content, supporting efficient navigation for users of assistive technologies[16].

Criteria 3: Colors (Perceivability) states that information must not rely on color alone. Text and text image must meet a minimum contrast ratio of 4.5:1 to remain readable for users with low vision or color vision deficiencies[16].

Criteria 4: Multimedia (Perceivability) requires synchronized alternatives for audio and video, such as transcripts, captions, and audio descriptions. Media players must also be keyboard-operable[16].

Criteria 5: Tables (Understandability) requires data tables to be structured so that relationships between headers and data cells are programmatically identifiable using elements such as `caption`, `scope`, `headers`, and `id`[16].

Criteria 6: Links (Understandability) states that the link text must clearly describe the purpose of the link. Vague expressions such as "click here" do not comply[16].

Criteria 8: Mandatory Document Structure (Robustness) requires a valid document type, a descriptive `<title>`, declared language, and semantic HTML elements (headings, lists, emphasis) to ensure compatibility with assistive technologies[16].

4 Assistive and Adaptive Technologies: Solutions for Digital Inclusion

Assistive technology is an umbrella term for products and related systems and services that help maintain or improve an individual's functioning, enabling their health, well-being, inclusion, and participation [17]. They act as a bridge between the user and the standard computing system. Adaptive technology is a subcategory of assistive technology specifically designed for people with disabilities, such as screen readers or Braille displays, whereas assistive technology can benefit a wider audience [18].

4.1 Solutions for Sensory Impairments

4.1.1 Visual Impairments

A wide range of assistive technologies convert visual information into other modalities for users with visual impairments.

Screen Readers: Software such as JAWS, NVDA, and VoiceOver read aloud text and

describe interface elements, enabling blind users to access digital content [19].

Screen Magnifiers: Applications like ZoomText and built-in OS magnifiers enlarge portions of the screen for users with low vision [20].

Refreshable Braille Displays: Hardware devices convert on-screen text into tactile Braille characters, providing an essential input/output channel for blind users [21].



Figure 6: Example of a refreshable Braille display.

Voice Assistants and Speech Recognition: Systems such as Siri, Google Assistant, and dictation tools support hands-free control and command execution [22].

Accessibility Settings: Operating systems provide built-in features such as high-contrast themes, enlarged cursors, large text, and color filters to assist users with low vision or color blindness [23].

4.1.2 Hearing Impairments

For users with hearing disabilities, solutions focus on presenting auditory information visually.

Captions and Subtitles: Provide text alternatives to spoken content in videos, as required under WCAG guidelines [7].

Transcripts: Full-text transcripts of audio recordings support comprehension [24].

Visual Alerts: System sounds can be replaced with flashing screen cues or pop-ups [25].

Speech-to-Text Engines: Real-time transcription tools convert speech to text during meetings or conversations [9, 22].

4.2 Solutions for Physical and Motor Impairments

These technologies provide alternative pathways for interacting with a computer.

Input Bandwidth Spectrum. The range of input efficiency, or "bandwidth," in human-computer interaction varies significantly, from the simple binary input of a single switch to the high-speed, parallel input required for CAD design and touch typing. Understanding this spectrum is crucial for selecting and designing appropriate assistive technologies for users with motor limitations [26].

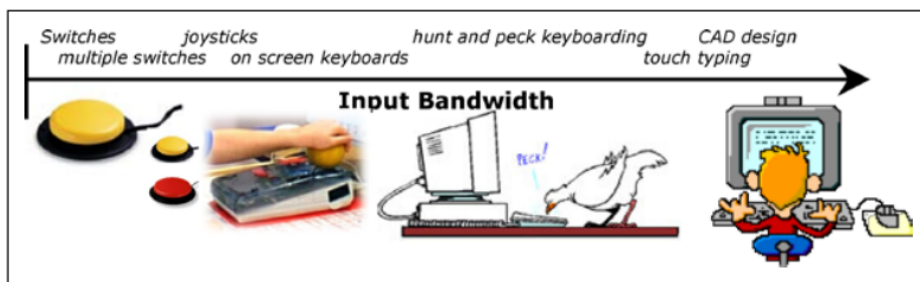


Figure 7: The input bandwidth spectrum for computer interaction, ranging from low bandwidth switches to high bandwidth touch typing.

Note. Adapted from “Switch access to technology — A comprehensive Guide,” by D. Colven & S. Judge (2006).

4.2.1 Alternative Pointing Devices

Tools such as trackballs, joysticks, head pointers, and adaptive mice reduce the need for fine motor movement [27].

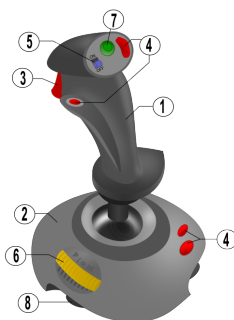


Figure 8: Diagram of a computer joystick components: 1. stick, 2. base, 3. trigger, 4. extra buttons, 5. autofire switch, 6. throttle, 7. hat switch (POV hat), 8. suction cups.

Note. From “Joystick,” by Wikipedia contributors, n.d., Wikipedia. Retrieved from <https://en.wikipedia.org/wiki/Joystick>. Copyright under Creative Commons.



Figure 9: The Kensington Expert Mouse Trackball.

Note. Diagram of a trackball alternative pointing device. From “Trackball,” by Wikipedia contributors, n.d., Wikipedia. Retrieved from <https://en.wikipedia.org/wiki/Trackball>. Copyright under Creative Commons.



Figure 10: The Adjustable Head Stylus, an example of a head pointer for hands-free tablet access.

Note. Image of the Adjustable Head Stylus. Source: RehabMart. Retrieved from <https://www.rehabmart.com/product/adjustable-head-stylus-40513.html?srsltid=AfmB0opM1cb546n-2WCLH6rFv7CGVDK-LGPFb506tqkT-4c2JrFVt00B>.

4.2.2 Eye-Tracking (Oculometry)

Screen-based eye-gaze systems detect where the user is looking and allow them to “click” by dwelling or blinking. These are crucial for individuals with severe motor impairments [28].

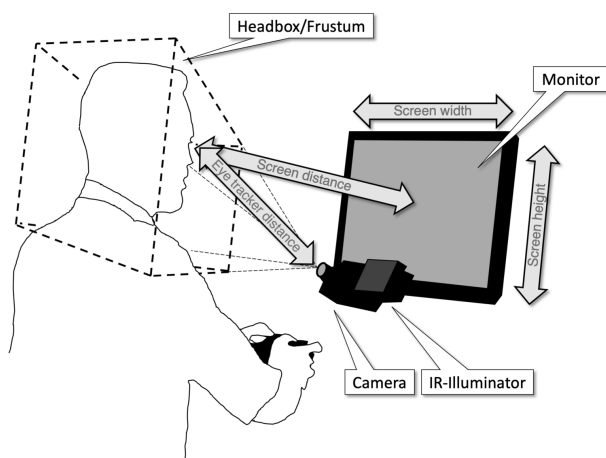


Figure 11: Example of a head-boxed eye-tracking setup.

Note. Adapted from “*RETRACTED ARTICLE: Eye tracking: empirical foundations for a minimal reporting guideline*,” retrieved from <https://link.springer.com/article/10.3758/s13428-021-01762-8>.

4.2.3 Switch Access

Single or multi-switch devices allow users to navigate interfaces through scanning mechanisms, activated by any reliable movement (hand, knee, sip-and-puff, etc.) [29].



Figure 12: A standard single switch, used as a primary input device for users with limited motor control.

Note. Adapted from “Switch access to technology — A comprehensive Guide,” by D. Colven & S. Judge (2006).

4.2.4 Speech Input

Voice-control systems allow users to operate computers and dictate text entirely by speech [30].

4.2.5 Adapted Keyboards

These include virtual on-screen keyboards, enlarged physical keyboards, and ergonomic layouts. Software features like Sticky Keys and Filter Keys enhance accessibility [31].



Figure 13: An adapted keyboard with a non-QWERTY layout.

Note. From “Keyboard and mouse alternatives and adaptations,” by AbilityNet, n.d., AbilityNet.

Retrieved from

<https://abilitynet.org.uk/factsheets/keyboard-and-mouse-alternatives-and-adaptations>.

Reprinted with permission.

4.2.6 Brain–Computer Interfaces (BCI)

Emerging BCI technology directly translates neural signals into computer commands, offering new possibilities for users with profound motor limitations [32].

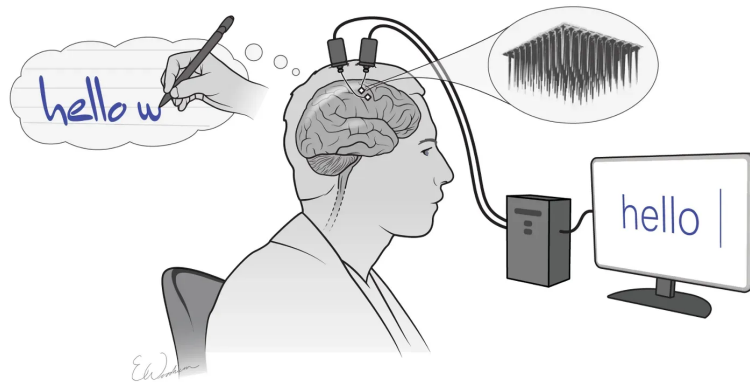


Figure 14: Conceptual diagram of a Brain-Computer Interface (BCI) for typing.

Note. From “Brain-Computer Interface Smashes Previous Record for Typing Speed,” by **Emily Waltz**, June 1, 2021, *IEEE Spectrum*. Retrieved from <https://spectrum.ieee.org/braincomputer-interface-smashes-previous-record-for-typing-speed>. Reprinted with permission.

4.3 Solutions for Cognitive Barriers

Fewer dedicated assistive technologies target cognitive disabilities directly. Accessible design practices based on Universal Design principles are the primary mechanism for reducing cognitive load [1, 12].

Simplified Interfaces: Clean layouts, reduced distractions, and strong visual hierarchy help users focus on essential content.

Predictable Navigation: Consistent menu placement, labels, and interaction patterns make it easier for users to understand and navigate interfaces.

Plain Language: Using simple, direct language reduces cognitive effort and improves comprehension.

Visual Aids and Icons: Complementing text with familiar icons supports understanding and memory.

Customizable Features: Adjustable text size, spacing, contrast, and animation settings allow users to tailor the interface to their needs [32].

Dyslexia-Friendly Fonts: Fonts such as OpenDyslexic or Sylexiad are specifically designed to enhance readability [11, 33].

5 Digital Accessibility in Algeria: Context and Gaps

Algeria signed the United Nations Convention on the Rights of Persons with Disabilities (UNCRPD) on 30 March 2007 and ratified it on 4 December 2009 [6]. The country also has a domestic law (Law 02/09, 8 May 2002) for the protection and promotion of persons with disabilities. Estimates indicate that Algeria has approximately 2 million persons living with disabilities, with physical impairments being the most common type [34, 35].

A 2021 study analysing national survey data (MICS 2006 and 2013) reported a disability prevalence of approximately 1.7% in 2013 and projects that the number of people with disabilities will increase by 2038 [4]. According to the same study, older adults (60+), males, and certain regions are more vulnerable to disability.

Despite these legal and demographic facts, there is no publicly available evidence that Algeria has enacted and enforced a comprehensive national regulation requiring public-sector websites, mobile applications, or other digital services to comply with web accessibility standards such as WCAG. A review of Arab region ICT accessibility legislation shows that no Arab country — including Algeria — currently enforces mandatory web-accessibility requirements [36]. Several sources note that, in practice, inclusion for persons with disabilities remains incomplete: infrastructure is often inadequate, and public services (physical and digital) are inconsistently accessible [37].

Thus, while Algeria has both a recognized legal framework and a substantial population with disabilities — with projections indicating growing disability prevalence — systematic implementation of digital accessibility appears lacking. There is no evidence of country-wide audits, accessibility enforcement mechanisms, or widespread compliance with international ICT accessibility guidelines.

6 Conclusion

This literature review has established the key connections between ergonomics, accessibility, and Human-Computer Interaction (HCI) for people with disabilities, with a focus on Algeria. It shows that barriers related to vision, hearing, movement, and cognition can significantly limit access to digital technologies, and that international standards such as ISO 9241-210, WCAG, and RGAA provide clear guidance for designing inclusive interfaces. Furthermore, a range of assistive and adaptive technologies exists to help users effectively overcome these barriers.

However, in Algeria, despite the presence of laws and international commitments, digital accessibility remains limited. Many public services and digital platforms are still difficult to use, and accessibility standards are not consistently enforced.

Therefore, to achieve genuine digital inclusion, Algeria must implement accessibility standards across digital services, support the development and use of assistive technologies, and promote digital literacy for both users and developers. By combining these efforts, the country can create a digital environment that is more usable, fair, and accessible for all citizens.

References

- [1] M. A. Salman. “Accessibility and usability challenges in human–computer interaction design”. In: *Journal of Humanities Research* 4.1 (2024), pp. 8–18.
- [2] Y. Fatima, S. Yadav, and A. Deharkar. “Human-Computer Interaction (HCI)”. In: *International Research Journal of Modernization in Engineering Technology and Science* 6.5 (2024).
- [3] C. Stephanidis et al. *Unified Interface Development: Tools for Constructing Accessible and Usable User Interfaces*. 1997.
- [4] K. Rachedi and S. Bouledras. “The prevalence of disability in Algeria: Reality and prospects”. In: *Algerian Journal of Relevant Field* (2021).
- [5] Author. “Accessibility in HCI: Analyzing inclusive design principles and techniques for ensuring accessibility to interactive systems for users with disabilities”. In: *Human-Computer Interaction Perspectives* 2.1 (2024), pp. 1–10.
- [6] United Nations. *Convention on the Rights of Persons with Disabilities (CRPD)*. 2006. URL: <https://www.un.org/development/desa/disabilities/convention-on-the-rights-of-persons-with-disabilities.html>.
- [7] World Wide Web Consortium (W3C). *Web Content Accessibility Guidelines (WCAG) 2.1*. 2018.
- [8] World Health Organization. *World report on vision*. 2021.
- [9] World Health Organization. *World report on hearing*. 2020.
- [10] J. P. Bigham et al. *Challenges and opportunities for accessibility*. 2017.
- [11] L. Rello and R. Baeza-Yates. “Good fonts for dyslexia”. In: *International Journal of Human–Computer Studies* 75 (2017), pp. 58–69.
- [12] A. Seffah and E. Metzker. *The foundations of usability engineering*. 2009.
- [13] International Organization for Standardization. *ISO 9241-210: Ergonomics of human–system interaction*. 2019.
- [14] World Wide Web Consortium (W3C). *Web Content Accessibility Guidelines (WCAG) 2.2*. 2023.
- [15] M. Halpin. *What are the levels of WCAG compliance?* Recite Me. URL: <https://reciteme.com/news/what-are-the-levels-of-wcag-compliance/>.
- [16] République Française. *Référentiel Général d’Amélioration de l’Accessibilité (RGAA) V4*. 2018.
- [17] World Health Organization. *Assistive technology factsheet*. 2018.

- [18] University of Wisconsin–Madison. *What are adaptive technologies?* 2025. URL: <https://website/path>.
- [19] WebAIM. *Screen Reader User Survey #9*. 2023.
- [20] American Foundation for the Blind. *Low Vision Assistive Technology*. 2022.
- [21] World Blind Union. *Braille Technology and Access to Information*. 2021.
- [22] A. M. S. Pradhan, K. Mehta, and L. Findlater. “Accessibility of voice assistants for users with disabilities”. In: *Proceedings of the ASSETS Conference*. 2018.
- [23] Microsoft. *Windows Accessibility Features Guide*. 2023.
- [24] U.S. Access Board. *Information and Communication Technology (ICT) Final Standards and Guidelines (Section 508)*. 2017.
- [25] World Wide Web Consortium (W3C). *WCAG 2.1 – Non-audio Notifications*. 2018.
- [26] A. M. Cook and J. M. Polgar. *Assistive Technologies: Principles and Practice*. 5th ed. Elsevier, 2020.
- [27] P. Majaranta and A. Bulling. “Eye tracking and eye-based human–computer interaction”. In: *Advances in Physiological Computing*. Springer, 2014, pp. 39–65.
- [28] H. H. Koester. *Switch Access to Technology*. RESNA, 2017.
- [29] D. Fitts and M. Heath. “Voice interaction for motor impairment: A review”. In: *Journal of Accessibility and Design* 1.1 (2019), pp. 45–60.
- [30] AbilityNet. *Keyboard and mouse alternatives and adaptations*. URL: <https://abilitynet.org.uk/factsheets/keyboard-and-mouse-alternatives-and-adaptations>.
- [31] J. R. Wolpaw and E. W. Wolpaw. *Brain–Computer Interfaces: Principles and Practice*. Oxford University Press, 2012.
- [32] A. Seffah et al. “Usability measurement and metrics: A practitioner’s guide”. In: *Software Quality Journal* 14.2 (2006), pp. 159–178.
- [33] British Dyslexia Association. *Guidelines for Dyslexia-Friendly Text*. 2022.
- [34] African Disability Rights Yearbook. *Country Report: Algeria*. 2021. URL: <https://www.adry.up.ac.za/section-b-2021/algeria>.
- [35] Handicap International. *Right to social protection for persons with disabilities in Algeria*. 2016.
- [36] Plateforme Arabe d’Inclusion Numérique. *Review of ICT Accessibility Legislation in the Arab Region*. 2023.

- [37] Le Courrier de l'Atlas. *Algérie: Le handicap peine à sortir d'une vision de charité*. 2023. URL: <https://www.lecourrierdelatlas.com/algerie-le-handicap-peine-a-sortir-d-une-vision-de-charite-20749>.